DA02 Values, names and types

Week 1, Meet the data | Session 2 of 24 | Python for Data Analysis

Today's question. What kind of thing is each part of a sale?

Objective. By the end: you can store numbers and text under names, and say what type each one is.

> WHERE THIS IS USED
> A till keeps a product name as text, a price as a number and a date as a date. If the types get mixed up, the totals go wrong. Which one is a phone number: text or a number?

Before this session: DA01 Open a notebook and read your first table. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
value	One piece of information, such as 2000 or Juice.	One entry on a receipt.
integer	A whole number with no decimal part.	Counting 3 books.
float	A number with a decimal part.	A price of 2500.5 shillings.
string	Text inside quotation marks.	A name written on a label.
variable	A name that points to a value.	A label on a box of stock.
type	The kind of value: integer, float or string.	Knowing whether a box holds pens or juice.

Block A. Names point to values

A shop sells 3 Juice bottles at 2000 shillings each. The equals sign stores a value under a name. The right side is worked out first.

Predict. Predict the three lines the cell will print. Write your guess on paper first.

Your guess, written before you run it

```
price = 2000
units = 3
revenue = price * units
print(revenue)
print(type(price))
print(type("2000"))
```

Run it, then compare. The cell prints 6000, then <class 'int'>, then <class 'str'>. price and units are variables. 2000 is an integer. The quoted "2000" is a string, which is text. The equals sign stores the value on its right under the name on its left.

Investigate.

1. Why is type("2000") a string?
2. What does the equals sign do in revenue = price * units?

Modify. Change units to 5 and run the whole cell again.

Block B. Decimals and converting text

Prices can have decimals. Text that looks like a number has to be converted before Python can calculate with it. The words float and int convert text into those types.

Predict. Predict what each of the first three prints shows, and whether the last line will work. Write your guess on paper first.

Your guess, written before you run it

```
price = 2500.5
text_price = "2500.5"
print(type(price))
print(type(text_price))
print(price * 2)
print(float(text_price) * 2)
```

Run it, then compare. The cell prints <class 'float'>, <class 'str'>, 5001.0 and 5001.0. The quoted price is text. float() turns it into a number that can be multiplied.

Investigate.

1. Why does the answer 5001.0 keep a decimal point even though 2 is a whole number?
2. What would happen with float("two")?

Modify. Convert the text "2500.5" to a number with float() and multiply it by 4.

Make

Create a decimal price of 2500.5 and a quantity of 2. Print the total. Write one sentence on why "2500.5" with quotation marks behaves differently.

Save your work as DA02. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
NameError: name 'price' is not defined	Python has not run a line that creates that name in this session.	Run the cell that creates the name first, and check the spelling and capital letters.
TypeError: can't multiply sequence by non-int of type 'float'	You tried to multiply text by a decimal.	Convert the text with float() or remove the quotation marks.

Exit ticket

A variable is a name that points to a value, and the value stays until I store a new one.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.